

*Fatima Jinnah Women University, Rawalpindi*

**Lab #1 : GitHub Identity, Student Portal Registration, and  
Ubuntu Server Installation**



**Name:**

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**Registration Number:**

2024-BSE-10

**Course Code:**

SE-665

**Section:**

B

**GitHub Username:**

muneezaamber

***Department of Software Engineering(BSE-5)***

## Getting Started

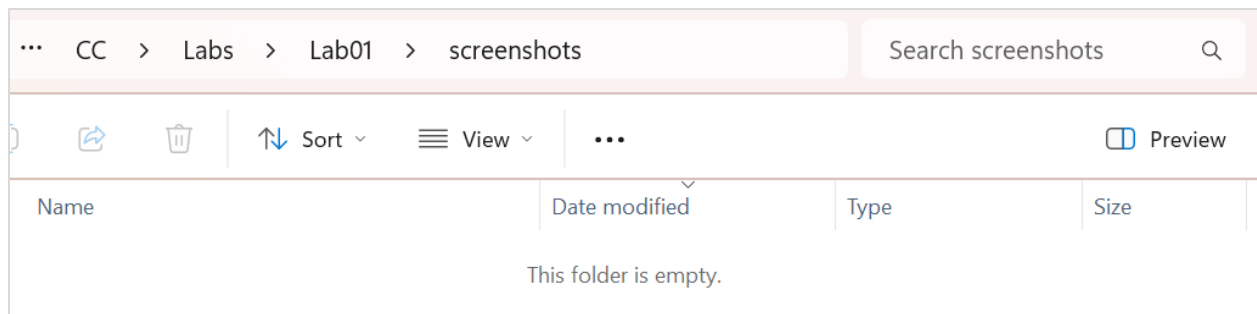


Figure 1: Local directory structure (Labs/Lab01/screenshots/) created for the Cloud Computing lab repository.

## Task 1: Create and Configure Your GitHub Account

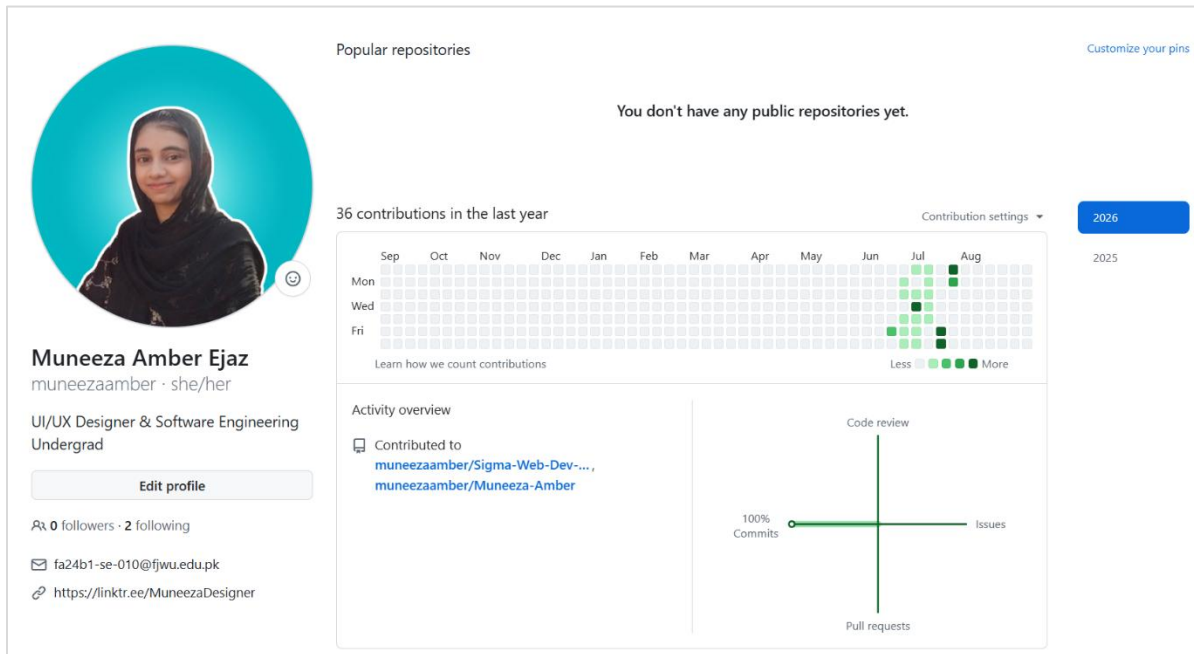


Figure 2: GitHub profile displaying the configured professional username and profile details.

### Public profile

**Name**

Muneeza Amber Ejaz

Figure 3: GitHub Public Profile settings showing the completed Name field with the actual name.

## Task 2: Register in the Student Marks Portal

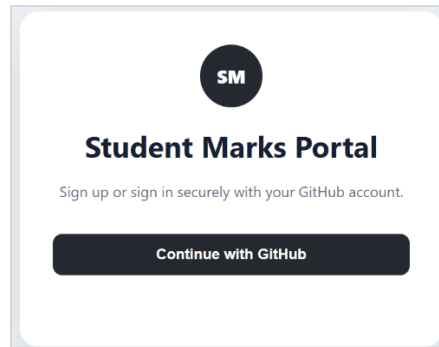


Figure 4: Student Marks Portal login page displaying the GitHub authentication option.

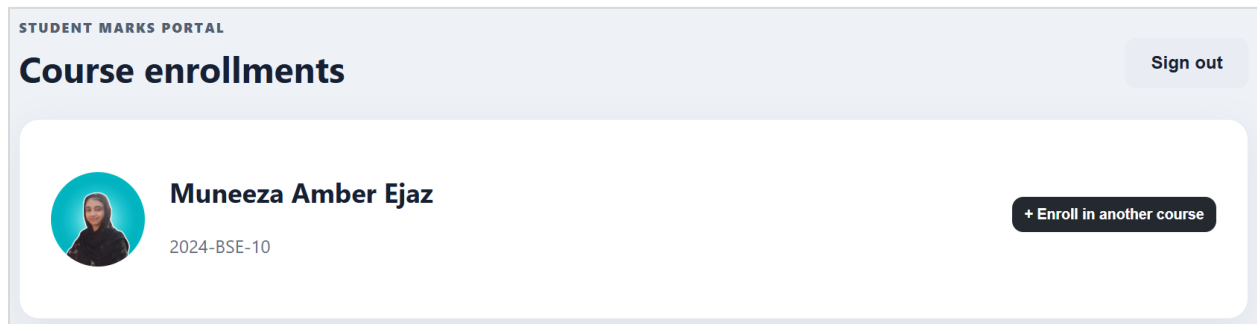


Figure 5: Course enrollments dashboard verifying the linked GitHub profile information and registration number.

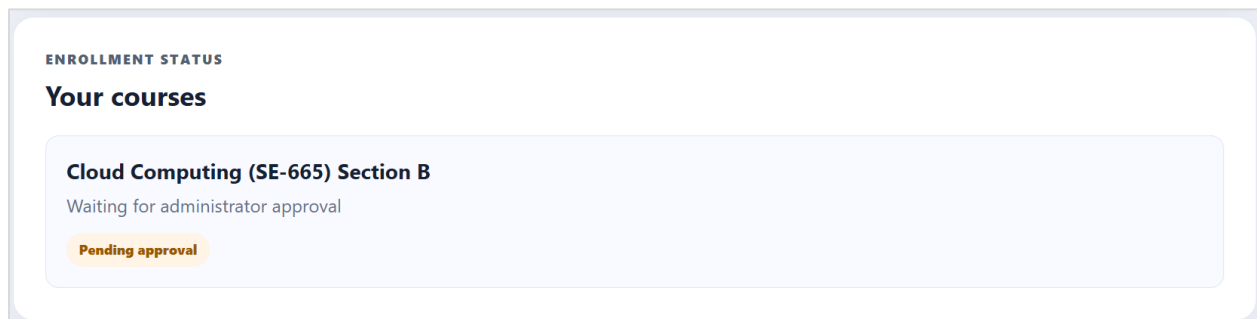


Figure 6: Enrollment status showing the submitted Cloud Computing course registration pending administrator approval.

## Task 3: Download VMware Workstation Pro and Ubuntu Server

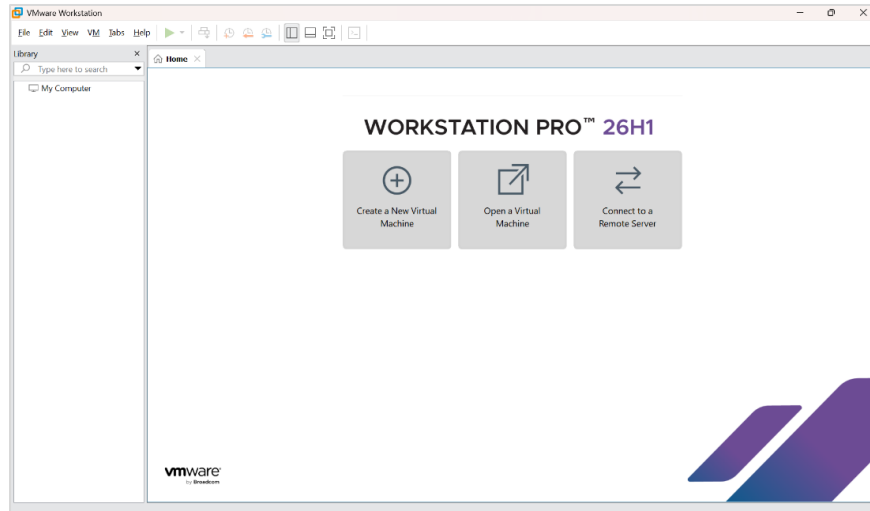


Figure 7: VMware Workstation Pro interface running successfully on the host machine.



Figure 8: The downloaded Ubuntu Server LTS ISO file ready for virtual machine installation.

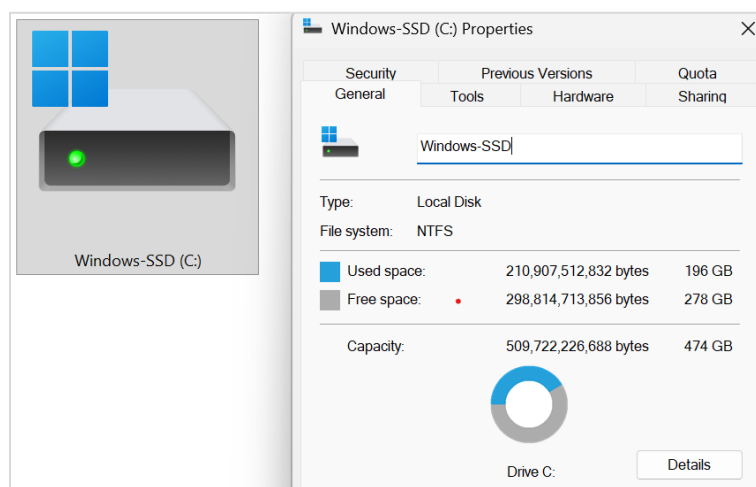


Figure 9: Local disk properties verifying sufficient available free space (over 20 GB) for the virtual machine.

## Task 4: Create the Ubuntu Server Virtual Machine



Figure 10: The New Virtual Machine Wizard initiated in VMware Workstation Pro.

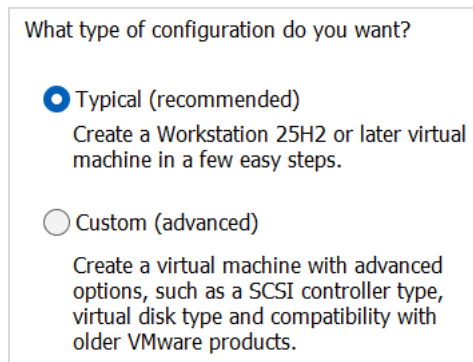


Figure 11: Selection of the Typical (recommended) configuration for the new virtual machine.

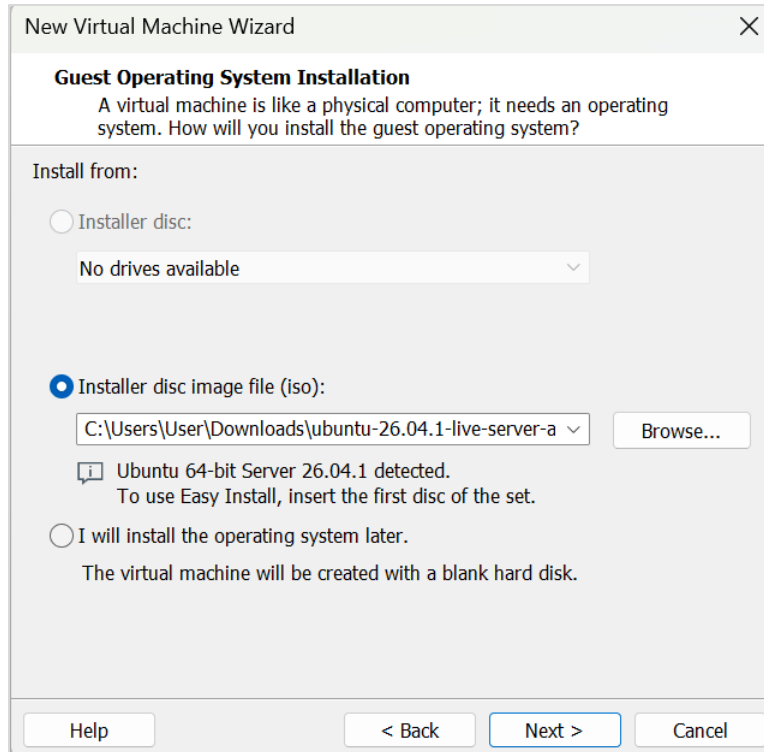


Figure 12: The downloaded Ubuntu Server ISO file selected as the installer disc image.

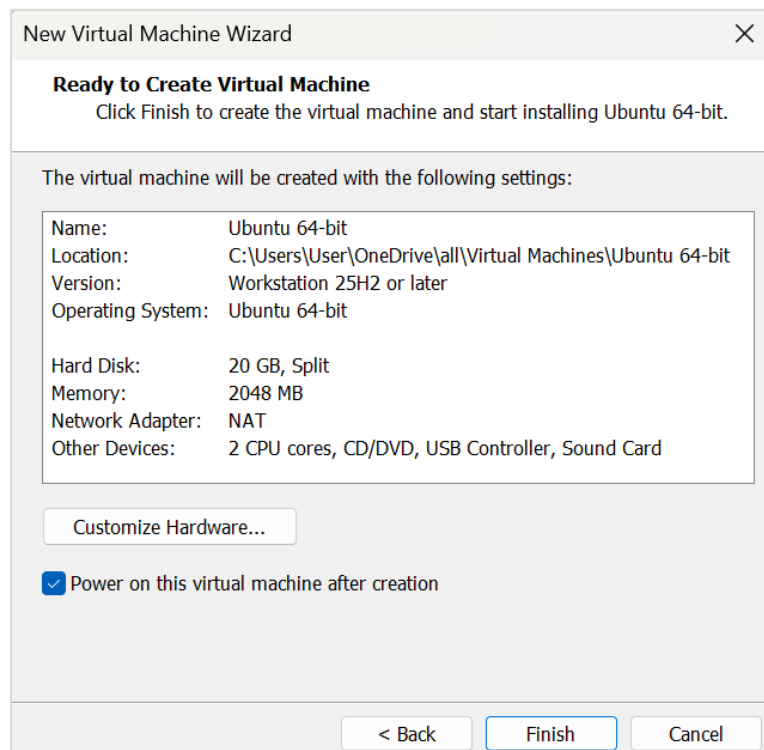


Figure 13: Final virtual machine configuration summary confirming the allocated resources.

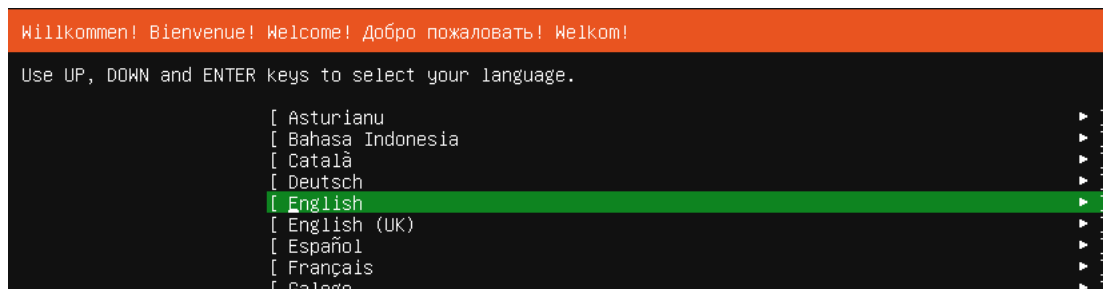
```

[DEPEND] Dependency failed for /subiquity_config.
[ OK ] Started Uncomplicated firewall.
[ OK ] Started Create list of required static device nodes for the current kernel.
[ OK ] Starting Create Static Device Nodes in /dev...
[ OK ] Mounted POSIX Message Queue File System.
[ OK ] Mounted Kernel Debug File System.
[ OK ] Started Remount Root and Kernel File Systems.
[ OK ] Starting Load/Save Random Seed...
[ OK ] Mounted Huge Pages File System.
[ OK ] Started Load/Save Random Seed.
[ OK ] Started Create Static Device Nodes in /dev.
[ OK ] Starting udev Kernel Device Manager...
[ OK ] Started Journal Service.
[ OK ] Starting Flush Journal to Persistent Storage...
[ OK ] Started udev Kernel Device Manager.
[ OK ] Started Load Kernel Modules.
[ OK ] Mounting Kernel Configuration File System...
[ OK ] Mounting FUSE Control File System...
[ OK ] Starting Apply Kernel Variables...
[ OK ] Started Flush Journal to Persistent Storage.
[ OK ] Mounted FUSE Control File System.
[ OK ] Mounted Kernel Configuration File System.
[ OK ] Started LVM2 metadata daemon.
[ OK ] Started udev Coldplug all Devices.
[ OK ] Started Monitoring of LVM2 mirrors, snapshots etc. using dmeventd or progress polling.
[ OK ] Started Apply Kernel Variables.

```

*Figure 14: The initial boot sequence of the Ubuntu Server installer.*

## Task 5: Choose the Installation Language



*Figure 15: Selection of English as the primary installation language for the server.*

## Task 6: Configure the Keyboard Layout



*Figure 16: Configuring the system keyboard layout to English (US).*



*Figure 17: Setting the appropriate keyboard variant to English (US).*

## Task 7: Choose the Ubuntu Server Installation Type

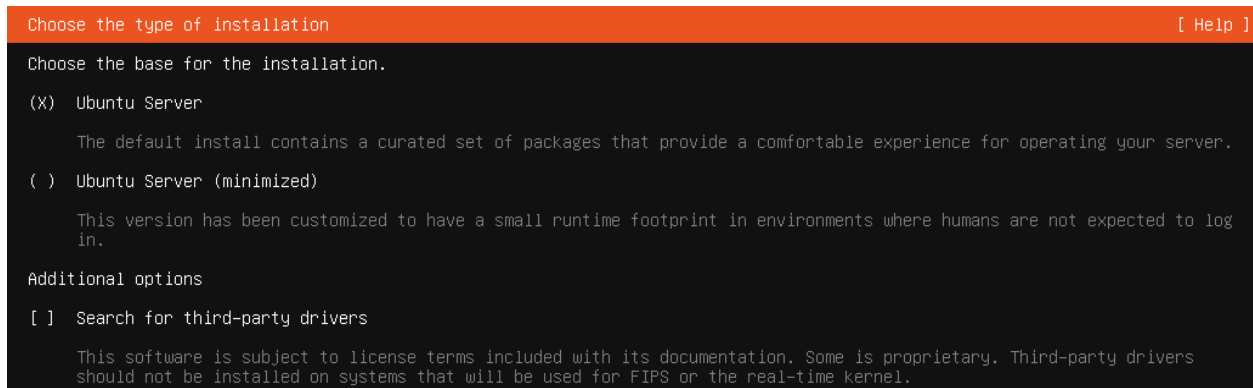


Figure 18: Choosing the standard Ubuntu Server installation base for the Cloud Computing lab.

## Task 8: Configure the Network

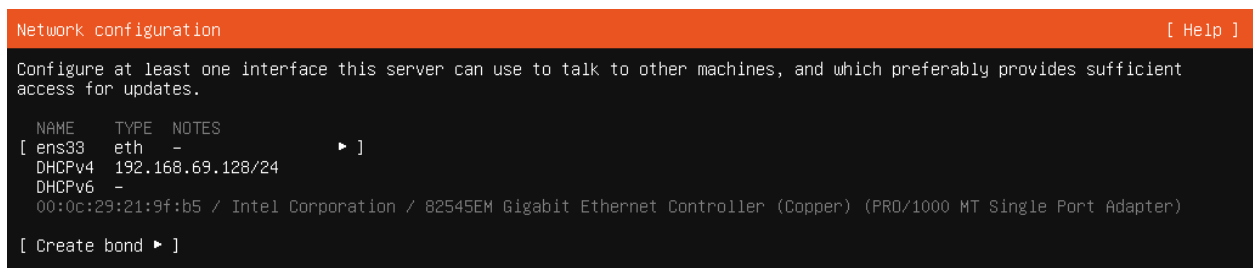


Figure 19: Network configuration screen displaying the detected ens33 ethernet interface.

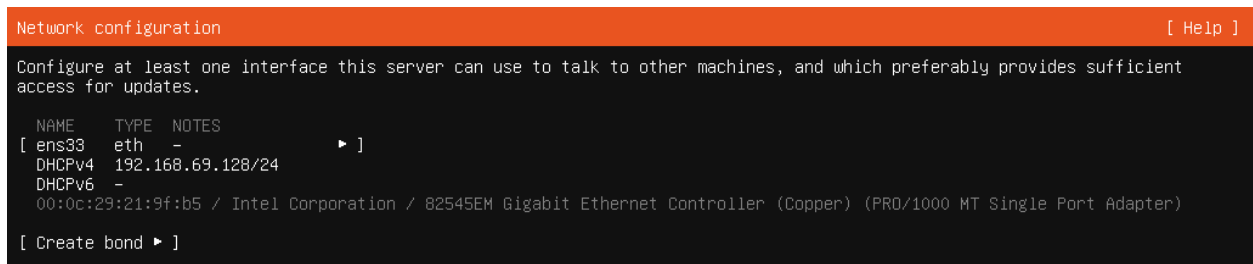


Figure 20: Confirmation of the IPv4 address assigned to the server automatically via DHCP.

## Task 9: Configure Storage

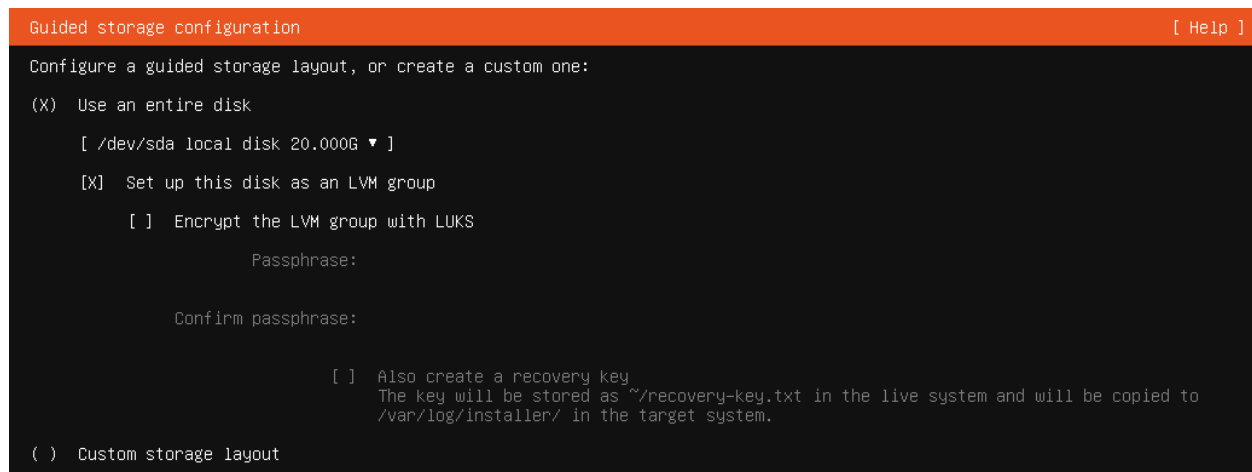


Figure 21: Selection of the guided storage option to utilize the entire VMware virtual disk.

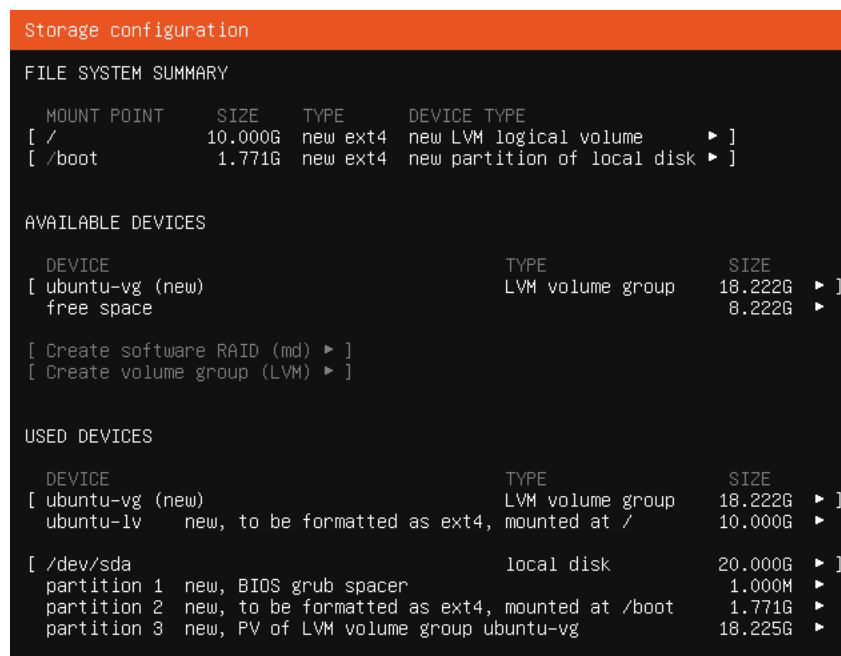


Figure 22: File system summary detailing the proposed storage configuration before formatting.

## Task 10: Select the Installation Disk

```

Storage configuration
FILE SYSTEM SUMMARY
  MOUNT POINT    SIZE    TYPE    DEVICE TYPE
[ /              10.000G new ext4 new LVM logical volume ▶ ]
[ /boot          1.771G new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES

  DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                        LVM volume group                    18.222G ▶ ]
  free space                             8.222G ▶ ]

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES

  DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                        LVM volume group                    18.222G ▶ ]
  ubuntu-lv    new, to be formatted as ext4, mounted at / 10.000G ▶ ]

[ /dev/sda                                local disk                          20.000G ▶ ]
  partition 1 new, BIOS grub spacer                        1.000M ▶ ]
  partition 2 new, to be formatted as ext4, mounted at /boot 1.771G ▶ ]
  partition 3 new, PV of LVM volume group ubuntu-vg         18.225G ▶ ]

```

Figure 23: Selection of the 20 GB virtual disk created specifically for the Ubuntu Server installation.

```

Storage configuration
FILE SYSTEM SUMMARY
  MOUNT POINT    SIZE    TYPE    DEVICE TYPE
[ /              10.000G new ext4 new LVM logical volume ▶ ]
[ /boot          1.771G new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES

  DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                        LVM volume group                    18.222G ▶ ]
  free space                             8.222G ▶ ]

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES

  DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                        LVM volume group                    18.222G ▶ ]
  ubuntu-lv    new, to be formatted as ext4, mounted at / 10.000G ▶ ]

[ /dev/sda                                local disk                          20.000G ▶ ]
  partition 1 new, BIOS grub spacer                        1.000M ▶ ]
  partition 2 new, to be formatted as ext4, mounted at /boot 1.771G ▶ ]
  partition 3 new, PV of LVM volume group ubuntu-vg         18.225G ▶ ]

```

Figure 24: Verification of the displayed disk size and device type to ensure the correct VMware virtual disk is targeted.

## Task 11: Review the Partitions

```

Storage configuration

FILE SYSTEM SUMMARY

MOUNT POINT      SIZE      TYPE      DEVICE TYPE
[ /               10.000G   new ext4   new LVM logical volume ▶ ]
[ /boot          1.771G   new ext4   new partition of local disk ▶ ]

AVAILABLE DEVICES

DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                    LVM volume group                    18.222G ▶ ]
free space                            8.222G ▶ ]

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES

DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                    LVM volume group                    18.222G ▶ ]
ubuntu-lv    new, to be formatted as ext4, mounted at / 10.000G ▶ ]
[ /dev/sda                            local disk                           20.000G ▶ ]
partition 1  new, BIOS grub spacer                        1.000M ▶
partition 2  new, to be formatted as ext4, mounted at /boot 1.771G ▶
partition 3  new, PV of LVM volume group ubuntu-vg          18.225G ▶

```

Figure 25: Review of the complete partition layout proposed by the installer to confirm it belongs to the virtual environment.

## Task 12: Confirm the Storage Changes

```

Confirm destructive action
Selecting Continue below will begin the installation process and
result in the loss of data on the disks selected to be formatted.

Are you sure you want to continue?

[ No ]
[ Continue ]

```

Figure 26: The destructive-action warning prompt triggered before formatting the selected virtual disk.

```

Confirm destructive action
Selecting Continue below will begin the installation process and
result in the loss of data on the disks selected to be formatted.

Are you sure you want to continue?

[ No ]
[ Continue ]

```

Figure 27: Confirmation to proceed with the storage changes and begin the installation process.

## Task 13: Configure the Ubuntu Profile

Your name:

*Figure 28: Entry of the complete actual name in the Ubuntu profile configuration.*

Your servers name:

*Figure 29: Configuration of the server name as 'ubuntu'.*

Pick a username:

*Figure 30: Entry of the exact GitHub username to serve as the system username.*

Profile configuration

[ Help ]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:   
The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

*Figure 31: Review of the fully completed Ubuntu profile setup before proceeding.*

## Task 14: Complete the Software Installation

```
Installing system [ Help ]

configuring partition: partition-1
configuring partition: partition-2
configuring format: format-0
configuring lvm_volgroup: lvm_volgroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
  writing install sources to disk
  running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmpv93brobi/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel
    setting up swap
    apply networking config
    writing etc/fstab
    configuring multipath
    updating packages on target system
    configuring pollinate user-agent on target
    configuring kernel crash dumps settings
    final kernel configuration
    configuring target system bootloader
    installing grub to target devices
    copying metadata from /cdrom
final system configuration
  calculating extra packages to install
  installing openssh-server
  retrieving openssh-server
```

*Figure 32: System installation process displaying the background package configuration.*

## **Task 15: Reboot and Verify the Ubuntu Identity**

```
Installation complete! [ Help ]

running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmpv93brobi/mount
configuring keyboard
  curtin command in-target
executing curtin install curthooks step
  curtin command install
    configuring installed system
      running 'curtin curthooks'
        curtin command curthooks
          configuring apt configuring apt
          installing missing packages
          configuring iscsi service
          configuring raid (mdadm) service
          configuring NVMe over TCP
          installing kernel
          setting up swap
          apply networking config
          writing etc/fstab
          configuring multipath
          updating packages on target system
          configuring pollinate user-agent on target
          configuring kernel crash dumps settings
          final kernel configuration
          configuring target system bootloader
          installing grub to target devices
          copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
retrieving openssh-server
  curtin command system-install
unpacking openssh-server
  curtin command system-install
configuring cloud-init
downloading and installing security updates
  curtin command in-target
restoring apt configuration
  curtin command in-target
subiquity/Late/run_builtin:
subiquity/Late/run_user_supplied:

[ View full log ]
[ Reboot Now ]
```

Figure 33: The installation complete screen prompting for a system reboot.

```
Ubuntu 26.04.1 LTS ubuntu tty1

Try contacting this VM's SSH server via 'ssh vsock%1411489717' from host.

ubuntu login: [ 25.244785] cloud-init[1159]: en_US.UTF-8... done
[ 25.244919] cloud-init[1159]: Generation complete.
```

Figure 34: The initial Ubuntu Server login prompt after a successful system boot.

```

ubuntu login: muneezaamber
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Sep 12 01:37:11 PM UTC 2026

System load:  0.01               Processes:    217
Usage of /:   50.6% of 9.75GB    Users logged in: 0
Memory usage: 22%               IPv4 address for ens33: 192.168.69.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

6 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

muneezaamber@ubuntu:~$ _

```

Figure 35: Successful local terminal login displaying the system welcome message.

```
muneezaamber@ubuntu:~$
```

Figure 36: Local terminal prompt verifying the required username and server identity.

## Task 16: Find the Ubuntu Server IP Address

```

muneezaamber@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:21:9f:b5 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c29219fb5
    inet 192.168.69.129/24 metric 100 brd 192.168.69.255 scope global dynamic ens33
        valid_lft 1309sec preferred_lft 1309sec
    inet6 fe80::20c:29ff:fe21:9fb5/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

```

Figure 37: Execution of the ip addr command to retrieve network interface details.

```

muneezaamber@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:21:9f:b5 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29219fb5
    inet 192.168.69.129/24 metric 100 brd 192.168.69.255 scope global dynamic ens33
        valid_lft 1309sec preferred_lft 1309sec
    inet6 fe80::20c:29ff:fe21:9fb5/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

```

Figure 38: Identification of the assigned IPv4 address for the active network interface.

## Task 17: Connect to Ubuntu Server from Windows

```

C:\Users\User>ssh muneezaamber@192.168.69.129

```

Figure 39: Execution of the remote SSH connection command from the Windows Command Prompt.

```

Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.69.129' (ED25519) to the list of known hosts.

```

Figure 40: Confirmation and acceptance of the Ubuntu server's security fingerprint.

```

Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Sat Sep 12 02:24:47 PM UTC 2026

System load:  0.0           Processes:            221
Usage of /:   50.7% of 9.75GB Users logged in:          0
Memory usage: 22%          IPv4 address for ens33: 192.168.69.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

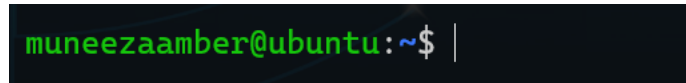
6 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

muneezaamber@ubuntu:~$

```

Figure 41: Successful remote SSH login displaying the Ubuntu Server welcome message.

A screenshot of a terminal window with a black background. The prompt 'muneezaamber@ubuntu:~\$' is displayed in a green monospace font, followed by a vertical cursor bar.

*Figure 42: Remote SSH terminal prompt verifying the user identity from the Windows host.*